MediNova Nexus

PVT. LTD.

Smart Care, Connected Life

SMART HOSPITAL MANAGEMENT SYSTEM

PROJECT REPORT

SUBMITTED TO:

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COMPANY NAME:

MediNova Nexus Pvt. Ltd.

GUIDED BY:

Dr NIBEDITA ADHIKARI

DATE:

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SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title:

Smart Hospital Management System

Company Name: MediNova Nexus Pvt. Ltd.

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1. INTRODUCTION

1.1 Purpose

This document describes the software requirements for the Smart Hospital Management System developed by MediNova Nexus Pvt. Ltd. The system aims to digitize and automate hospital operations such as patient management, appointments, billing, and medical records.

1.2 Scope

The system will provide an integrated platform for hospitals to manage patients, doctors, appointments, prescriptions, and

billing efficiently. It improves healthcare service delivery and reduces manual workload.

1.3 Definitions

- HMS: Hospital Management System
- EHR: Electronic Health Record
- Admin: System administrator
- Patient: User seeking healthcare services

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2. OVERALL DESCRIPTION

2.1 Product Perspective

The system is a web-based application with optional mobile support. It connects patients, doctors, and administrators through a centralized platform.

2.2 Product Functions

- Patient registration and management

- Doctor scheduling and management
- Appointment booking system
- Electronic medical records (EMR)
- Billing and payment system
- Notification and alert system

2.3 User Classes

- Admin: Full system access
- Doctor: Manage appointments and patient records
- Patient: Book appointments and view reports
- Receptionist: Handle registrations and scheduling

2.4 Operating Environment

- Web browsers (Chrome, Edge)
- Mobile devices (Android/iOS optional)
- Backend server (Cloud/Local)

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3. SYSTEM FEATURES

3.1 Patient Management

- Register new patients
- Update patient details
- Maintain medical history

3.2 Doctor Management

- Add/manage doctor profiles
- Set availability schedule

3.3 Appointment System

- Book, cancel, reschedule appointments
- Real-time slot availability

3.4 Electronic Health Records (EHR)

- Store prescriptions and reports
- Access patient history

3.5 Billing System

- Generate invoices
- Payment tracking (cash/online)

3.6 Notification System

- Appointment reminders
- SMS/Email alerts

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4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interface

- Dashboard for each user role
- Simple, responsive UI
- Graphs and patient data visualization

4.2 Hardware Interface

- Computer systems
- Optional medical devices integration

4.3 Software Interface

- Database: MySQL
- Backend: Node.js / Python
- Frontend: HTML, CSS, JavaScript

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5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance

- System response time < 3 seconds
- Support multiple users simultaneously

5.2 Security

- User authentication (Login system)
- Role-based access control
- Data encryption

5.3 Reliability

- 99% uptime availability
- Automatic backup

5.4 Scalability

- Can support multiple hospitals
- Expandable database

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6. SYSTEM ARCHITECTURE

Client (User Interface) → Application Server → Database Server

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7. USE CASES

Use Case 1: Book Appointment

- Actor: Patient
- Description: Patient selects doctor and books slot

Use Case 2: Manage Patient Records

- **Actor: Doctor**
- **Description: Doctor views and updates medical records**

Use Case 3: Generate Bill

- **Actor: Admin**
- **Description: Generate and manage billing**

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8. DATABASE DESIGN

Tables:

- **Users (ID, Name, Role, Login)**
- **Patients (ID, Medical History)**
- **Doctors (ID, Specialization)**
- **Appointments (Date, Time, Status)**
- **Billing (Amount, Payment Status)**

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9. TESTING REQUIREMENTS

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing

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10. FUTURE ENHANCEMENTS

- AI-based disease prediction
- Telemedicine (video consultation)
- Mobile app version
- Integration with wearable health devices

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11. CONCLUSION

The Smart Hospital Management System by MediNova Nexus Pvt. Ltd. will enhance healthcare efficiency, reduce manual errors, and improve patient experience through digital transformation.

12. DETAILED FUNCTIONAL REQUIREMENTS

12.1 User Authentication

- Users must log in using username and password
- System shall support password encryption
- Invalid login attempts should be restricted

12.2 Patient Module

- Add new patient details
- View patient history
- Update patient records
- Delete patient data (Admin only)

12.3 Doctor Module

- Add/edit doctor details
- Assign specialization
- Manage availability schedule

12.4 Appointment Module

- Book appointment
- Cancel appointment
- Reschedule appointment
- View appointment history

12.5 Billing Module

- Generate bill automatically
- Include consultation + treatment charges
- Support multiple payment methods

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13. NON-FUNCTIONAL REQUIREMENTS (DETAILED)

13.1 Usability

- Easy-to-use interface
- Minimal training required

13.2 Maintainability

- Modular system design

- Easy to update features

13.3 Portability

- Works on different operating systems

- Accessible via browser

13.4 Availability

- System should be available 24/7

- Downtime < 1%

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14. DATA FLOW DESCRIPTION

1. User logs into system

2. System validates credentials

3. User performs operation (booking, billing, etc.)

4. Data stored in database

5. Output shown on dashboard

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15. SEQUENCE FLOW (APPOINTMENT)

1. Patient logs in

2. Selects doctor

3. Checks available slots

4. Books appointment

5. System confirms booking

6. Notification sent

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16. ACTIVITY FLOW

Start → Login → Dashboard → Select Function → Process → Save Data → Output → End

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17. RISK ANALYSIS

Risk	Impact	Solution
Server Failure	High	Backup server
Data Loss	High	Regular backup
Security Breach	High	Encryption & firewall
System Crash	Medium	Error handling

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18. ASSUMPTIONS

- Users have basic knowledge of computers
- Internet connection is available
- Hospital staff will use the system properly

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19. LIMITATIONS

- Requires internet connection
- Initial setup cost
- Training required for staff

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20. HARDWARE REQUIREMENTS

- Computer system (4GB RAM minimum)
- Server system
- Internet connectivity

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21. SOFTWARE REQUIREMENTS

- Operating System: Windows/Linux
- Backend: Node.js / Python
- Frontend: HTML, CSS, JavaScript
- Database: MySQL

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22. INSTALLATION REQUIREMENTS

- Install server software
- Configure database

- Deploy application
- Test system

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23. USER MANUAL (SHORT)

1. Login to system
2. Select role (Admin/Doctor/Patient)
3. Perform required action
4. Logout after use

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24. MAINTENANCE PLAN

- Regular system updates
- Database backup weekly
- Bug fixing and improvements

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25. GLOSSARY

- EHR: Electronic Health Record

- Admin: System controller

- Dashboard: Main user interface

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26. FINAL CONCLUSION

The Smart Hospital Management System developed by MediNova Nexus Pvt. Ltd. provides a complete digital solution for hospital operations. It enhances efficiency, reduces errors, and improves patient care through automation and centralized management.

FINAL END OF SRS DOCUMENT

End of Document